Azure AI

Azure AI Translator

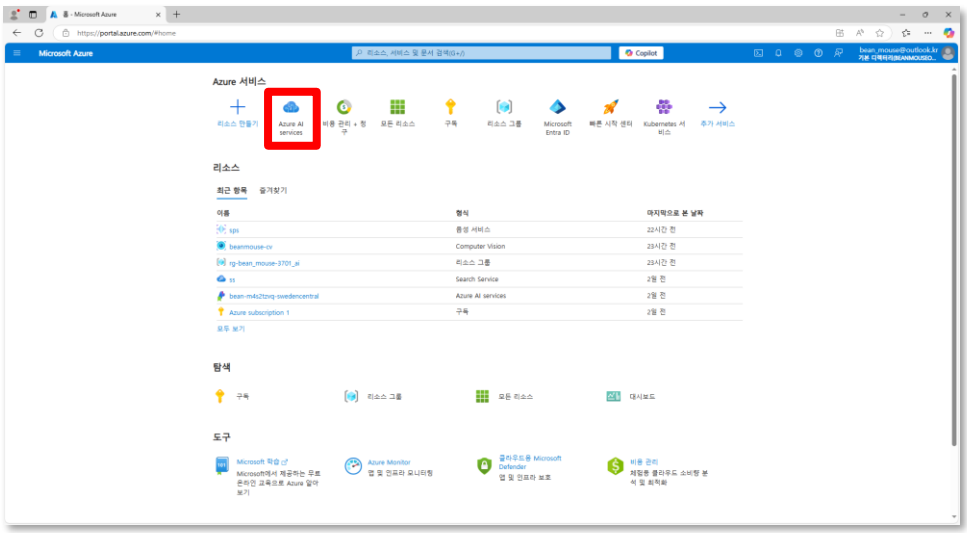
권기웅

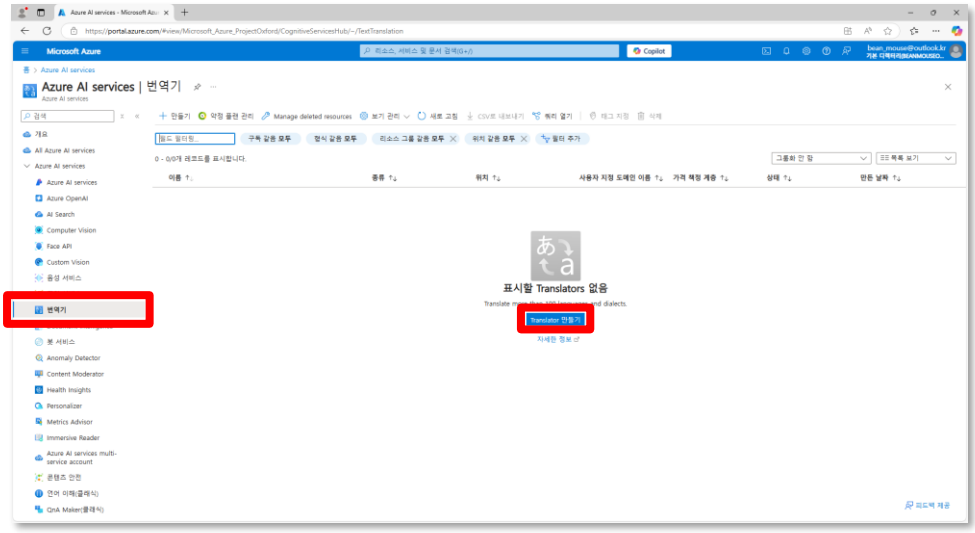
1. 환경설정

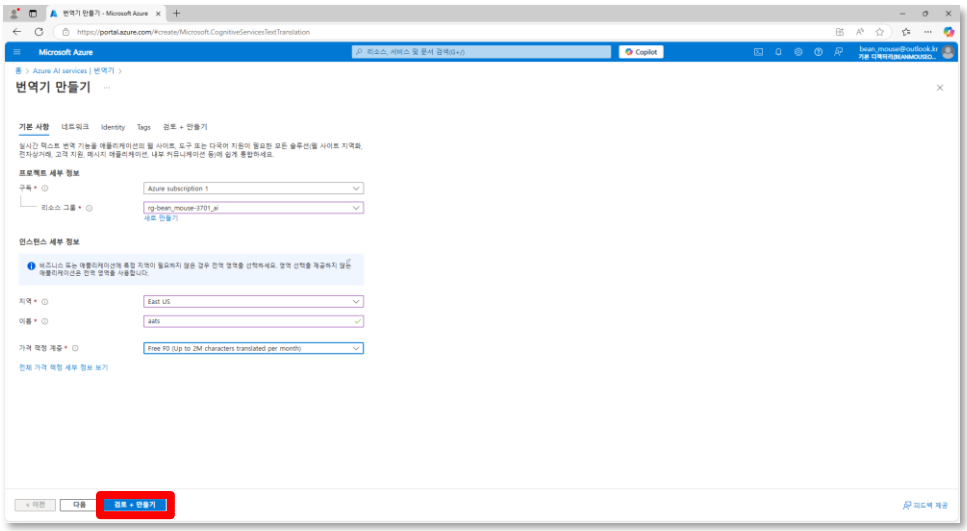
1.1 Azure AI Translator 리소스

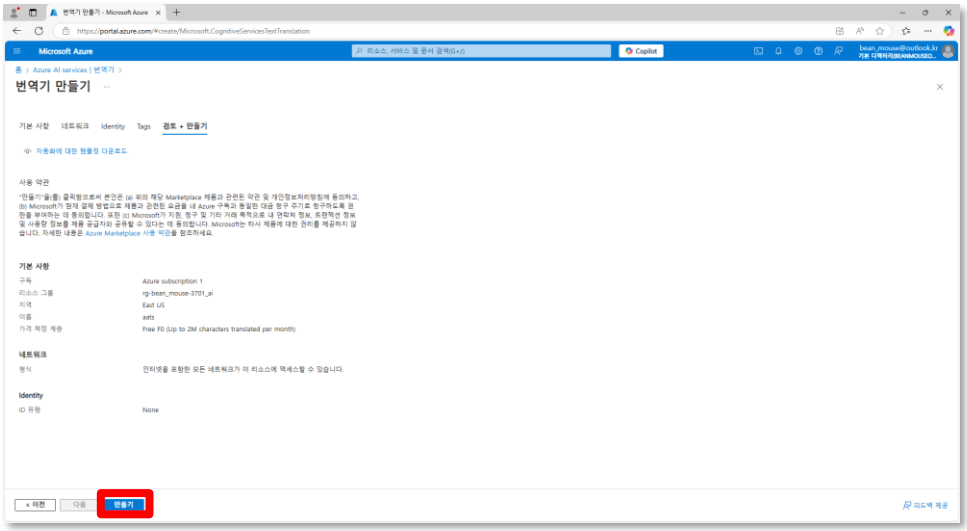
1.1.1 리소스 만들고

https://portal.azure.com 에서

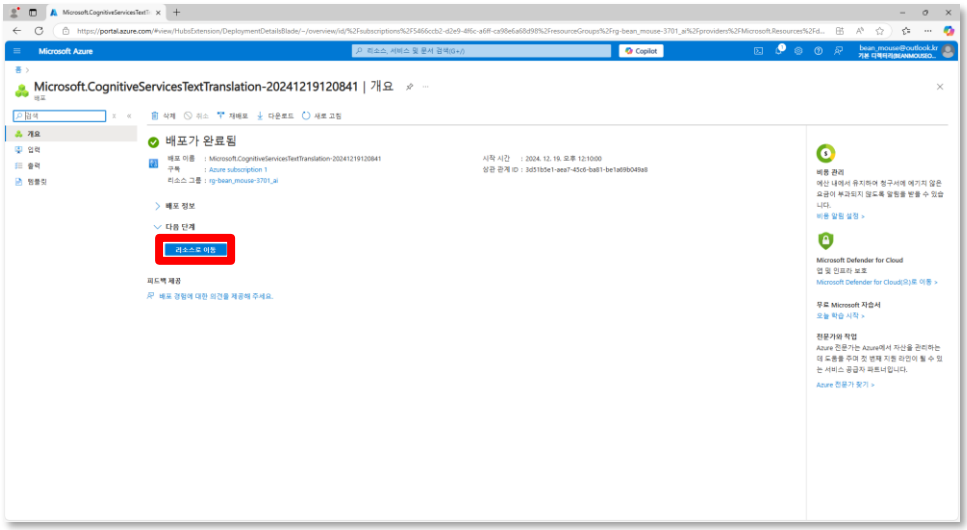


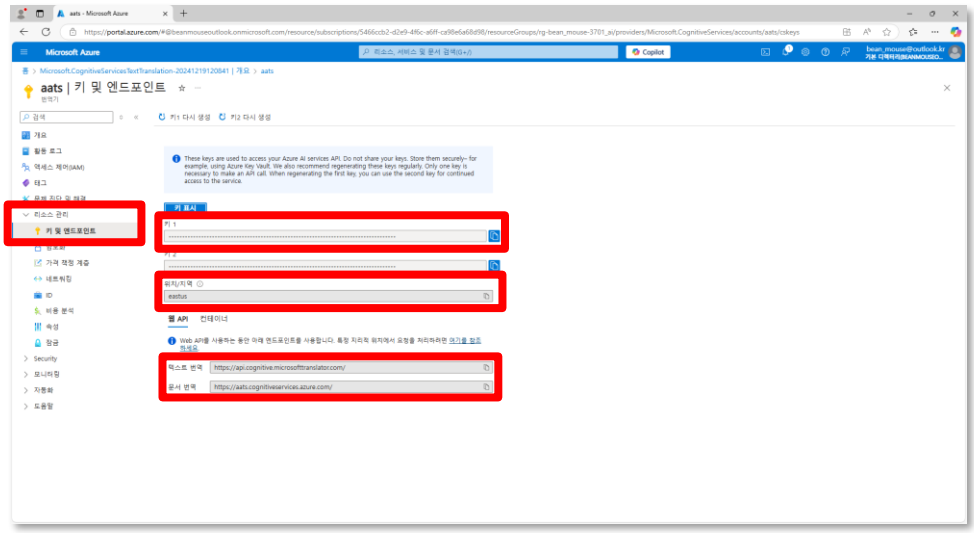



번역기 만들기 - Microsoft Azure
https://portal.azure.com/#create/Microsoft.CognitiveServicesTextTranslation
Microsoft Azure
Copilot
홈 > Azure AI services | 번역기 >
번역기 만들기
기본 사항
네트워크
Identity
Tags
검토 + 만들기
프로젝트 세부 정보
구독 *
Azure subscription 1
리소스 그룹 *
rg-bean_mouse-3701_ai
새로 만들기
인스턴스 세부 정보
지역 *
East US
이름 *
aafs
가격 책정 계층 *
Free F0 (Up to 2M characters translated per month)
전체 가격 책정 세부 정보 보기
< 이전
다음
검토 + 만들기
피드백 제공


번역기 만들기 - Microsoft Azure
https://portal.azure.com/#create/Microsoft.CognitiveServicesTextTranslation
Microsoft Azure
Copilot
홈 > Azure AI services | 번역기 >
번역기 만들기
기본 사항
네트워크
Identity
Tags
검토 + 만들기
자동화에 대한 템플릿 다운로드
사용 약관
기본 사항
구독
Azure subscription 1
리소스 그룹
rg-bean_mouse-3701_ai
지역
East US
이름
aafs
가격 책정 계층
Free F0 (Up to 2M characters translated per month)
네트워크
형식
인터넷을 포함한 모든 네트워크가 이 리소스에 액세스할 수 있습니다.
Identity
ID 유형
None
< 이전
다음
만들기
피드백 제공

1.2 키, 지역, 엔드포인트 챙기기

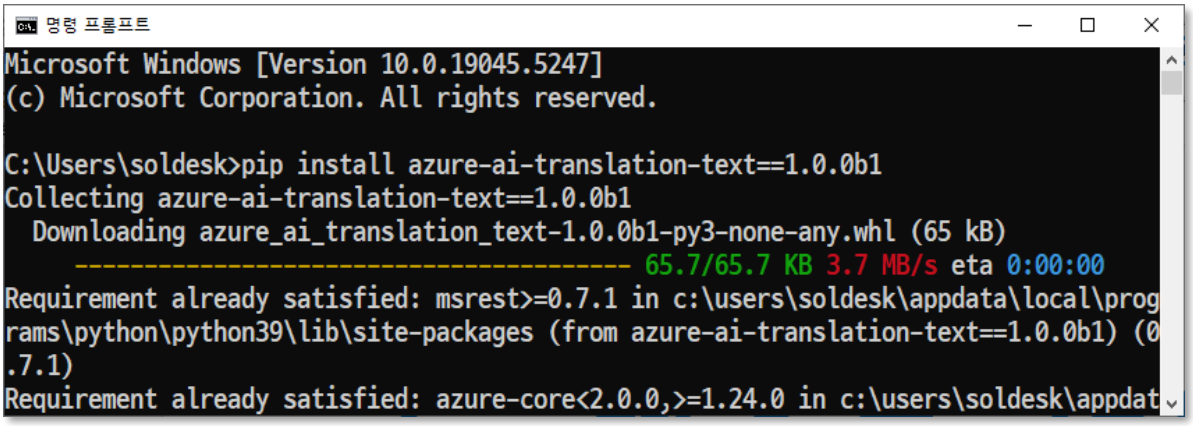




2. 텍스트 번역

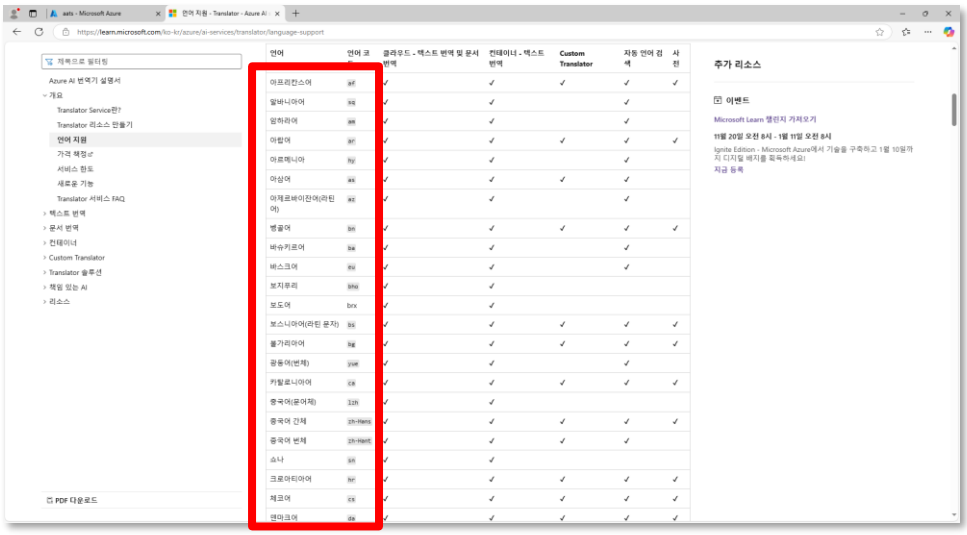
2.1 Python 라이브러리 설치

```
pip install azure-ai-translation-text==1.0.0b1
```



2.2 언어 코드 확인

https://learn.microsoft.com/ko-kr/azure/ai-services/translator/language-support



2.3 Python 프로그램

```python
# -*- coding:utf-8 -*-
from azure.ai.translation.text import TextTranslationClient, TranslatorCredential
from azure.ai.translation.text.models import InputTextItem

key = "FbjdSv1k1tZFeKcqW3o8adfe54E27jVCKkkvI4SldQVcrBZ3oPXFJQQJ99ALACYeBjFXJ3w3AAAbACOGT1IX"  # API키
endpoint = "https://api.cognitive.microsofttranslator.com/"  # 엔드포인트
region = "eastus"  # 지역

credential = TranslatorCredential(key, region)
text_translator = TextTranslationClient(endpoint=endpoint, credential=credential)

target_languages = ["en", "ja", "zh-Hans"]
input_text_elements = [InputTextItem(text="단무지, 피클 반입금지")]

response = text_translator.translate(content=input_text_elements, to=target_languages)
translation = response[0]
print("원래 문장 :", translation["detectedLanguage"])

for translated_text in translation.translations:
    print(translated_text.to + " : " + translated_text.text)
```

```
PS D:\Kwon\workspaceMS>  d:; cd 'd:\Kwon\work
ions\ms-python.debugpy-2024.14.0-win32-x64\bu
원래 문장 : {'language': 'ko', 'score': 1.0}
en : Pickles are not allowed
ja : ピクルスは許可されていません
zh-Hans : 不允许使用泡菜
PS D:\Kwon\workspaceMS>
```

3. 문서 번역

3.1 무료로는 불가능해서

요금제를 바꿔야 하는데

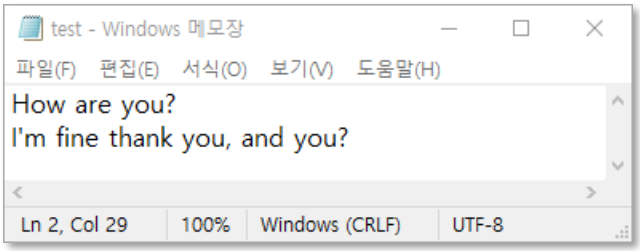
3.2 그냥 텍스트 번역으로 구현하는게

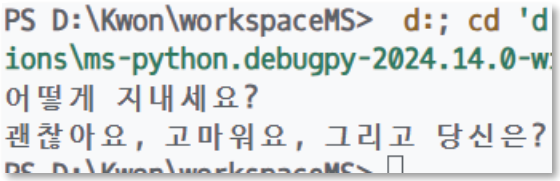
낫지 않을까

3.3 txt 파일

3.3.1 Python 프로그램

```python
# -*- coding:utf-8 -*-
from azure.ai.translation.text import TextTranslationClient, TranslatorCredential
from azure.ai.translation.text.models import InputTextItem

key = "FbjdSv1k1tZFeKcqW3o8adfe54E27jVCKkkvI4SldQVcrBZ3oPXFJQQJ99ALACYeBjFXJ3w3AAAbACOGT1IX"
endpoint = "https://api.cognitive.microsofttranslator.com/"
region = "eastus"

credential = TranslatorCredential(key, region)
text_translator = TextTranslationClient(endpoint=endpoint, credential=credential)
target_languages = ["ko"]

file_path = "D:/Kwon/자원/etc/test.txt"
f = open(file_path, "r", encoding="utf-8")
input_text_elements = []
for line in f.readlines():
    line = line.replace("\n", "")
    input_text_elements.append(InputTextItem(text=line))
f.close()

response = text_translator.translate(content=input_text_elements, to=target_languages)
for r in response:
    for translated_text in r.translations:
        print(translated_text.text)
```





3.4 docx 파일

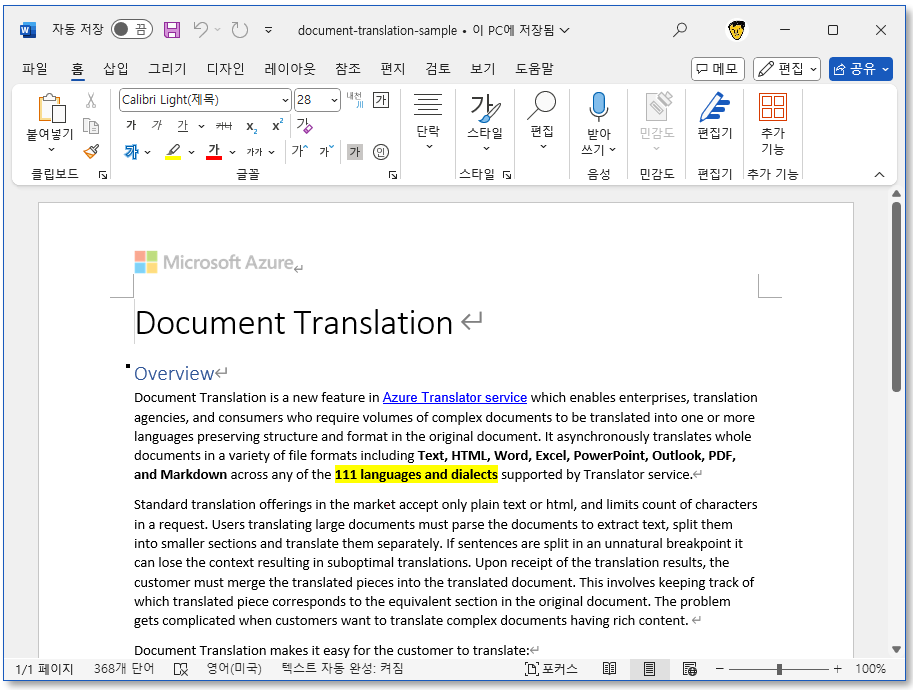
3.4.1 라이브러리 설치

```
pip install python-docx
```

```
명령 프롬프트
Microsoft Windows [Version 10.0.19045.5247]
(c) Microsoft Corporation. All rights reserved.

C:\Users\soldesk>pip install python-docx
Collecting python-docx
  Downloading python_docx-1.1.2-py3-none-any.whl (244 kB)
     ---------------------------------------- 244.3/244.3 KB ? eta 0:00:00
Collecting lxml>=3.1.0
  Downloading lxml-5.3.0-cp39-cp39-win_amd64.whl (3.8 MB)
     ---------------------------------------- 3.8/3.8 MB 15.2 MB/s eta 0:00:00
Requirement already satisfied: typing-extensions>=4.9.0 in c:\users\soldesk\appdata
```

3.4.2 Python 프로그램

```python
# -*- coding:utf-8 -*-
from azure.ai.translation.text import TextTranslationClient, TranslatorCredential
from azure.ai.translation.text.models import InputTextItem
from docx import Document

key = "FbjdSv1k1tZFeKcqW3o8adfe54E27jVCKkkvI4SldQVcrBZ3oPXFJQQJ99ALACYeBjFXJ3w3AAAbACOGT1IX"
endpoint = "https://api.cognitive.microsofttranslator.com/"
region = "eastus"

credential = TranslatorCredential(key, region)
text_translator = TextTranslationClient(endpoint=endpoint, credential=credential)
target_languages = ["ko"]

file_path = "D:/Kwon/document-translation-sample.docx"
input_text_elements = []
document = Document(file_path)
for p in document.paragraphs:
    input_text_elements.append(InputTextItem(text=p.text))

response = text_translator.translate(content=input_text_elements, to=target_languages)
for r in response:
    for translated_text in r.translations:
        print(translated_text.text)
```

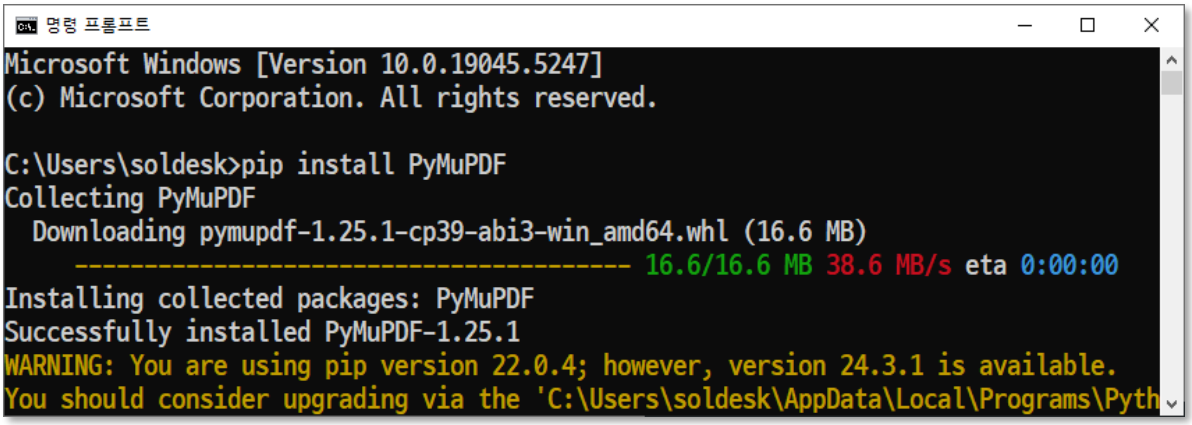



```
PS D:\Kwon\workspaceMS>  d:; cd 'd:\Kwon\workspaceMS'; & 'c:\Users\soldesk\AppData\Local\
ions\ms-python.debugpy-2024.14.0-win32-x64\bundled\libs\debugpy\adapter/../..\debugpy\lau
문서 번역
개요
문서 번역은 Azure Translator 서비스의 새로운 기능으로, 복잡한 문서를 하나 이상의 언어로 
을 유지하면서 번역할 수 있도록 합니다. Translator 서비스에서 지원하는 111개 언어 및 방언
 포함한 다양한 파일 형식의 전체 문서를 비동기식으로 번역합니다.
```

3.5 pdf 파일

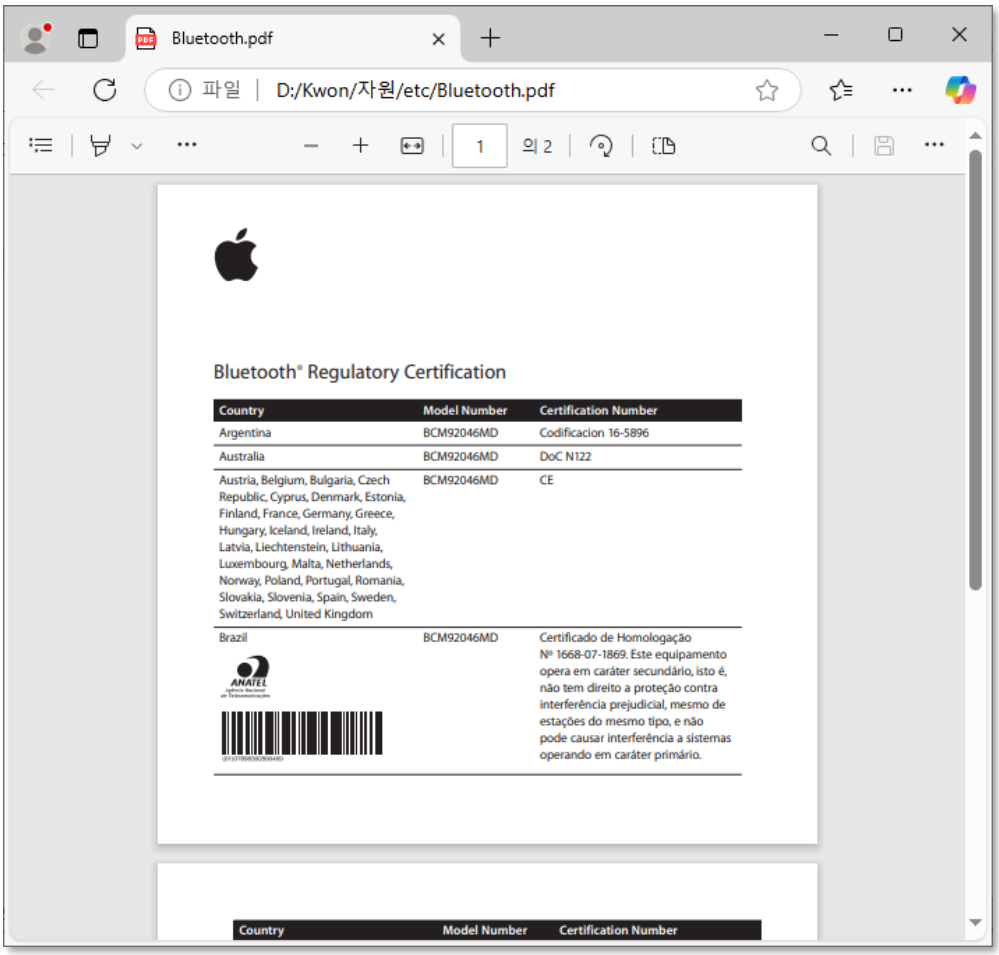
3.5.1 라이브러리 설치

```
pip install PyMuPDF
```



3.5.2 Python 프로그램

```python
# -*- coding:utf-8 -*-
from azure.ai.translation.text import TextTranslationClient, TranslatorCredential
from azure.ai.translation.text.models import InputTextItem
import fitz

key = "FbjdSv1k1tZFeKcqW3o8adfe54E27jVCKkkvI4SldQVcrBZ3oPXFJQQJ99ALACYeBjFXJ3w3AAAbACOGT1IX"
endpoint = "https://api.cognitive.microsofttranslator.com/"
region = "eastus"

credential = TranslatorCredential(key, region)
text_translator = TextTranslationClient(endpoint=endpoint, credential=credential)
target_languages = ["ko"]

file_path = "D:/Kwon/자원/etc/Bluetooth.pdf"
input_text_elements = []
doc = fitz.open(file_path)
for page in doc:
    input_text_elements.append(InputTextItem(text=page.get_text()))
doc.close()

response = text_translator.translate(content=input_text_elements, to=target_languages)
for r in response:
    for translated_text in r.translations:
        print(translated_text.text)
```



Bluetooth.pdf

파일 | D:/Kwon/자원/etc/Bluetooth.pdf

1 의 2

Bluetooth® Regulatory Certification

Country	Model Number	Certification Number
Argentina	BCM92046MD	Codificacion 16-5896
Australia	BCM92046MD	DoC N122
Austria, Belgium, Bulgaria, Czech Republic, Cyprus, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Liechtenstein, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, Sweden, Switzerland, United Kingdom	BCM92046MD	CE
Brazil	BCM92046MD	Certificado de Homologação Nº 1668-07-1869. Este equipamento opera em caráter secundário, isto é, não tem direito a proteção contra interferência prejudicial, mesmo de estações do mesmo tipo, e não pode causar interferência a sistemas operando em caráter primário.

Country	Model Number	Certification Number

```
PS D:\Kwon\workspaceMS>
ions\ms-python.debugpy-2

Bluetooth® 규정 인증
나라
모델 번호
인증번호
```